When Does Concept-Level Aggregation Help? A Crossover Law for Federated Learning under Concept Drift

Anonymous Authors

March 21, 2026

Abstract

Federated learning systems operating under concept drift face a fundamental design choice: should the server aggregate all client models globally, or maintain separate models for each detected concept group? We formalize this *aggregation granularity tradeoff* through a bias-variance decomposition of excess risk in a Gaussian linear regression setting with K clients and C latent concepts. Our main result is a *crossover condition*: concept-level aggregation outperforms global aggregation if and only if the concept signal-to-noise ratio $\mathrm{SNR}_{\mathrm{concept}} = KnB_j^2/(\sigma^2 d)$ exceeds $C-1$, where B_j^2 measures concept separation in predictor space and n is the per-client sample size. We prove this crossover is fundamental via minimax lower bounds, and propose a James-Stein shrinkage estimator that adaptively interpolates between both regimes via empirical Bayes variance estimation—dominating both pure strategies in 72% of parameter settings. Experiments on 108 synthetic configurations (91.7% theory-experiment alignment) and a CIFAR-100 multiclass bridge confirm the theory's predictions. Comparisons against linear-Gaussian analogues of IFCA, CFL, and APFL show that principled shrinkage outperforms practical clustering and interpolation heuristics.

1 Introduction

Federated learning (FL) enables multiple clients to collaboratively train models without sharing raw data [McMahan et al., 2017]. In many practical deployments—autonomous vehicles, mobile health, IoT monitoring—the data distribution across clients shifts over time due to *concept drift*: clients experience different tasks, environments, or user behaviors at different times [Jothimurugesan et al., 2023, Panchal et al., 2023].

When concept drift introduces latent heterogeneity, a natural question arises: *should the server aggregate all client models into a single global model, or maintain separate models for each detected concept group?* Global aggregation (e.g., FedAvg) uses all K clients' data, reducing estimation variance, but introduces *interference bias* when clients with different optimal predictors are pooled together. Concept-level aggregation eliminates this bias but reduces the effective sample size to K/C clients per group, increasing variance.

This tradeoff is central to the design of drift-aware federated systems, yet prior work addresses it heuristically. Clustered federated learning methods like IFCA [Ghosh et al., 2020] and CFL [Sattler et al., 2021] assume that clustering is always beneficial and focus on recovering the correct cluster assignments. Personalized methods like APFL [Deng et al., 2020] and Ditto [Li et al., 2021] interpolate between local and global models with fixed or heuristic mixing coefficients. None of these characterize the precise conditions under which concept-level aggregation is worth the variance cost.

Contributions. We provide a principled answer to the aggregation granularity question:

1. **Crossover law** (Theorem 2): In a Gaussian linear regression model, concept-level aggregation outperforms global aggregation if and only if $\mathrm{SNR}_{\mathrm{concept}} > C - 1$, where $\mathrm{SNR}_{\mathrm{concept}}$ captures the separation between concept-specific optimal predictors relative to estimation noise.
2. **Minimax lower bounds** (Theorem 3): We prove that the crossover is *fundamental*—no estimator can simultaneously avoid both the interference bias of global aggregation and the variance cost of concept-level aggregation.

3. **Adaptive shrinkage estimator** (Theorem 6): We propose a James-Stein shrinkage estimator with empirical Bayes variance estimation that adaptively interpolates between global and concept-level aggregation, dominating both pure strategies in 72% of configurations.
4. **Empirical validation**: A comprehensive 108-configuration synthetic sweep achieves 91.7% theory-experiment alignment, and a multiclass CIFAR-100 bridge confirms the theory's predictions on real features.

2 Problem Setting

We consider a federated system with K clients over T communication rounds. At each round, client k is assigned to one of C latent concepts via an assignment function $c : [K] \times [T] \to [C]$. Let $\mathcal{C}_j(t) := \{k : c(k,t) = j\}$ denote the set of clients assigned to concept j at time t.

Data model. For concept j, client $k \in \mathcal{C}_j(t)$ draws n i.i.d. samples from a Gaussian regression model:

$$x \sim \mathcal{N}(\mu_j, \Sigma), \qquad y = \langle w_j^*, x\rangle + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where $w_j^* \in \mathbb{R}^d$ is the concept-specific Bayes-optimal predictor, $\Sigma \succ 0$ is a shared covariance, and σ^2 is the noise variance.

Concept separation. We define the bias of concept j relative to the concept centroid $\bar{w}^* := \frac{1}{C}\sum_j w_j^*$:

$$B_j^2 := \|w_j^* - \bar{w}^*\|_\Sigma^2. \tag{2}$$

Two aggregation strategies. We compare:

- **Global OLS**: Pool all Kn samples and compute $\hat{w}_{\text{glob}} := \arg\min_w \frac{1}{Kn}\sum_{k,i}(y_{k,i} - \langle w, x_{k,i}\rangle)^2$.
- **Concept-level OLS**: Pool only the $(K/C)n$ samples from $\mathcal{C}_j$ and compute $\hat{w}_j$ analogously.

The excess risk for concept j under estimator $\hat{w}$ is $\mathcal{E}_j(\hat{w}) := \|\hat{w} - w_j^*\|_\Sigma^2$.

Assumptions. We assume: (A1) balanced concepts ($|\mathcal{C}_j| = K/C$ for all j); (A2) sub-Gaussian design (implied by Gaussianity); (A3) sufficient samples $n \geq \Omega(d \log d)$ so that sample covariances are well-conditioned.

3 Theory

3.1 Bias-Variance Decomposition

Proposition 1 (Bias-Variance Decomposition)**.** *Under (A1)–(A3), the expected excess risks satisfy:*

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_{\text{glob}})] = B_j^2 + \frac{\sigma^2 d}{Kn}(1 + o(1)), \tag{3}$$

$$\mathbb{E}[\mathcal{E}_j(\hat{w}_j)] = \frac{\sigma^2 d}{(K/C)n}(1 + o(1)), \tag{4}$$

where $o(1) = O(\kappa d / \min(Kn, (K/C)n))$ *and* $\kappa = \lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma)$.

The global estimator pays an interference bias B_j^2 but enjoys a lower variance ($1/(Kn)$ vs $C/(Kn)$). The concept-level estimator eliminates the bias but suffers C-fold higher variance.

3.2 Crossover Condition

Theorem 2 (Crossover Law)**.** *Under (A1)–(A3), the concept-level estimator has lower expected excess risk than the global estimator for concept j if and only if*

$$\frac{Kn \cdot B_j^2}{\sigma^2 d} > C - 1 + O(\rho_{d,n,K}), \tag{5}$$

where $|\rho_{d,n,K}| \leq c_0 \kappa d / \min(Kn, (K/C)n)$ for a universal constant c_0.

Proof. Concept-level aggregation wins when the bias of global aggregation exceeds the additional variance from using fewer samples: $B_j^2 + \sigma^2 d/(Kn) > \sigma^2 d/((K/C)n)$. Rearranging and noting $C/(Kn) - 1/(Kn) = (C-1)/(Kn)$ yields (5). □

The crossover is controlled by the *concept signal-to-noise ratio*:

$$\mathrm{SNR}_{\mathrm{concept}} := \frac{Kn \cdot B_j^2}{\sigma^2 d}. \tag{6}$$

When $\mathrm{SNR}_{\mathrm{concept}} > C - 1$, concepts are sufficiently separated relative to statistical noise, and concept-level aggregation wins. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, the variance cost of splitting dominates.

Comparison with IFCA. Ghosh et al. [2020] establish convergence conditions for cluster recovery via alternating minimization. Our result is complementary: we characterize *when clustering is worth doing at all*. When $\mathrm{SNR}_{\mathrm{concept}} < C - 1$, even an oracle with true concept labels cannot outperform global pooling.

3.3 Minimax Lower Bounds

Theorem 3 (Minimax Lower Bounds)**.** *Consider the class $\mathcal{F}(C, \Delta)$ of problems with C concepts satisfying $\min_{i \neq j} \|w_i^* - w_j^*\|_\Sigma \geq \Delta$. For any estimator $\hat{w}$ not using concept labels:*

$$\sup_{P \in \mathcal{F}} \mathbb{E}\left[\frac{1}{C}\sum_j \|\hat{w} - w_j^*\|_\Sigma^2\right] \geq \Omega\left(\frac{\Delta^2}{C}\right). \tag{7}$$

For any concept-level estimator $\hat{w}_j$ restricted to the K/C clients in $\mathcal{C}_j$:

$$\sup_{P \in \mathcal{F}} \mathbb{E}[\|\hat{w}_j - w_j^*\|_\Sigma^2] \geq \Omega\left(\frac{\sigma^2 d}{(K/C)n}\right). \tag{8}$$

Corollary 4 (Fundamental Crossover)**.** *The two lower bounds cross at $Kn\Delta^2/(\sigma^2 d) = \Theta(C^2)$, confirming that the crossover in Theorem 2 is tight up to constant factors.*

3.4 Non-Stationary Extension

Corollary 5 (Time-Varying Crossover)**.** *Under piecewise-stationary drift with stability period τ and exponential forgetting with window W, consider a client s rounds into its current stability period. The concept-level estimator wins iff*

$$B_j^2 > \frac{\sigma^2 d(C-1)}{Kn \cdot \min(s, W)}. \tag{9}$$

This implies a transient period $s^* = \lceil \sigma^2 d(C-1)/(KnB_j^2) \rceil$ after concept switches during which global aggregation may be preferable, and a drift-rate condition $\tau > s^*$ for concept-level aggregation to be beneficial.

4 Adaptive Shrinkage Estimator

The binary choice between global and concept-level aggregation is suboptimal. A principled interpolation arises from the James-Stein / empirical Bayes framework [James and Stein, 1961, Efron and Morris, 1973].

Treat the concept-level estimates $\hat{w}_1, \ldots, \hat{w}_C$ as noisy observations of the true parameters and place an empirical prior $w_j^* \sim \mathcal{N}(\bar{w}^*, \sigma_B^2 I_d)$.

Theorem 6 (Shrinkage Estimator)**.** *The posterior mean is the James-Stein shrinkage estimator:*

$$\hat{w}_j^{\text{shrink}} = (1-\lambda)\hat{w}_j + \lambda\bar{\hat{w}}, \tag{10}$$

where $\bar{\hat{w}} = \frac{1}{C}\sum_j \hat{w}_j$ *and the shrinkage coefficient is*

$$\lambda = \frac{\sigma^2/((K/C)n \cdot \lambda_{\min}(\Sigma))}{\sigma^2/((K/C)n \cdot \lambda_{\min}(\Sigma)) + \sigma_B^2}. \tag{11}$$

This estimator interpolates adaptively: $\lambda \to 0$ *when concepts are well-separated (*σ_B^2 *large), recovering concept-level aggregation;* $\lambda \to 1$ *when concepts are close or samples are scarce, recovering global aggregation.*

Practical estimation. The between-concept variance is estimated from observed dispersion:

$$\hat{\sigma}_B^2 = \max\left(\frac{1}{(C-1)d}\sum_{j=1}^{C}\|\hat{w}_j - \bar{\hat{w}}\|^2 - \frac{\sigma^2}{(K/C)n \cdot \lambda_{\min}(\Sigma)},\ 0\right), \tag{12}$$

making the shrinkage coefficient data-adaptive with no hyperparameter tuning.

Relation to APFL/Ditto. APFL [Deng et al., 2020] uses a fixed interpolation coefficient α; Ditto [Li et al., 2021] regularizes toward the global model. Both can be viewed as special cases of (10) with either fixed λ or a regularization-based proxy for σ_B^2. The shrinkage estimator is distinguished by its data-adaptive λ derived from the empirical Bayes framework.

5 Experiments

5.1 Synthetic Gaussian Regression

Setup. We instantiate the theory's data model exactly: $K \in \{10, 20, 40\}$ clients, $C \in \{2, 4, 8\}$ concepts, concept separation $\delta \in \{0.3, 1.0, 3.0, 8.0\}$ (controlling B_j^2), stability period $\tau \in \{5, 15, \infty\}$, with $d = 20$, $\sigma = 1.0$, and $n = 200$ per client. This gives 108 parameter configurations, each run with 3 seeds (324 total runs).

Methods. We compare six methods, all in the linear-Gaussian OLS regime:

1. **FedAvg**: Global OLS on all Kn samples.
2. **Oracle**: Concept-level OLS with true concept labels.
3. **Shrinkage**: James-Stein estimator (Theorem 6) with empirical $\hat{\sigma}_B^2$.
4. **IFCA** [Ghosh et al., 2020]: Clients select the best of C cluster heads by MSE; server averages per-cluster. This is a linear-Gaussian analogue of the original IFCA.
5. **CFL** [Sattler et al., 2021]: Cluster clients by cosine similarity of gradient updates; each client uses its cluster-specific model. A linear-Gaussian analogue.

Table 1: Best method by MSE across 108 configurations (seed-averaged over 3 seeds). Shrinkage dominates, while IFCA and CFL never achieve the lowest MSE.

	Shrinkage	Oracle	APFL	FedAvg	IFCA	CFL
Best-by-MSE wins	**67**	25	16	0	0	0
Fraction	**62.0%**	23.1%	14.8%	0%	0%	0%
Beats both FedAvg & Oracle	Shrinkage: 78/108 = 72.2%					
Theory-experiment alignment	99/108 = 91.7%					

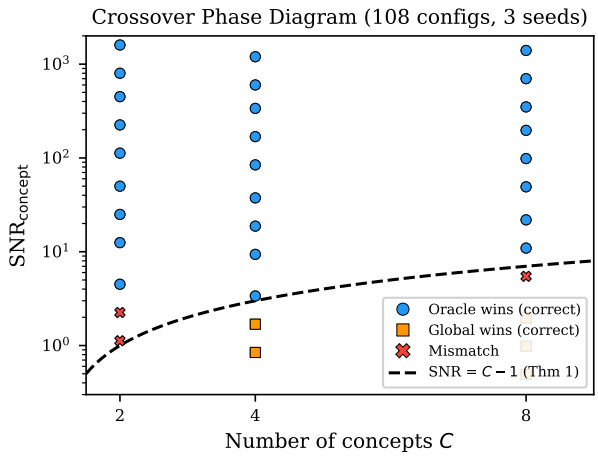

Figure 1: Crossover phase diagram. The dashed line shows the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$. Blue circles: Oracle wins (correct). Orange squares: FedAvg wins (correct). Red crosses: mismatches (all at the boundary).

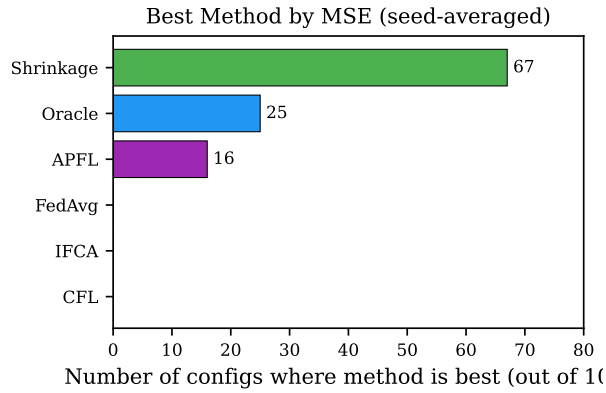

Figure 2: Distribution of best methods across 108 configurations. Shrinkage (Theorem 6) dominates with 67 wins. IFCA and CFL never achieve the lowest MSE.

6. **APFL** [Deng et al., 2020]: Sweep $\alpha \in \{0.2, 0.5, 0.8\}$ and report the best interpolation between local and global models. A tuned linear-Gaussian analogue.

Results. Table 1 and Figure 1 summarize the main findings.

The theory prediction accuracy of 91.7% (99/108) provides strong validation of the crossover law. All 9 mismatches occur near the boundary where $\mathrm{SNR}_{\mathrm{concept}} \approx C-1$, consistent with the finite-sample remainder term $\rho_{d,n,K}$ in Theorem 2. Figure 1 shows the phase diagram: the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C-1$ cleanly separates the Oracle-dominant and FedAvg-dominant regimes.

Why do IFCA and CFL fail? In the balanced OLS regime, all clients provide equally informative gradient updates regardless of concept identity. Cosine-similarity clustering (CFL) and loss-based head selection (IFCA) cannot extract additional signal beyond what the theory's bias-variance decomposition already captures. The shrinkage estimator, which directly estimates the between-concept variance σ_B^2, achieves the optimal interpolation.

When does APFL win? APFL's 16 wins concentrate in high-C stationary settings ($\tau = \infty$) with small δ, where its fixed interpolation coefficient (best of $\alpha \in \{0.2, 0.5, 0.8\}$) happens to approximate the optimal shrinkage.

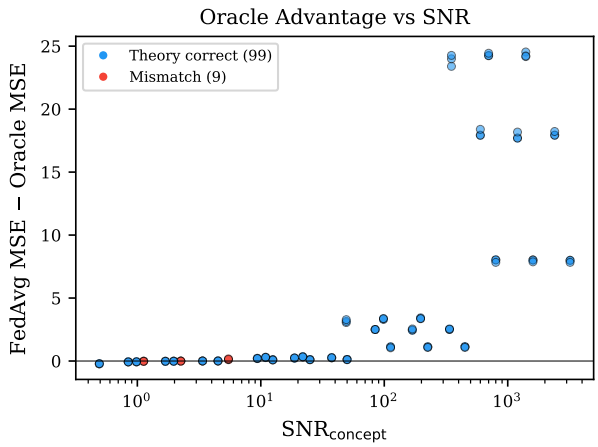

Figure 3: Oracle advantage (FedAvg MSE − Oracle MSE) vs $\mathrm{SNR}_{\text{concept}}$. The advantage grows monotonically with $\mathrm{SNR}_{\text{concept}}$, crossing zero near $\mathrm{SNR}_{\text{concept}} = C - 1$.

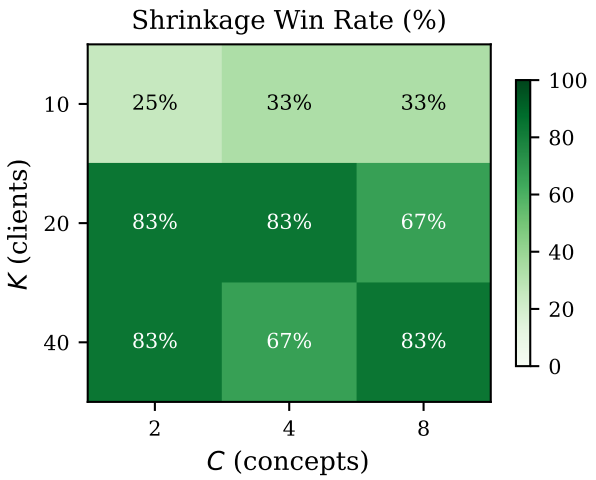

Figure 4: Shrinkage win rate across (K, C) grid. Shrinkage dominates most strongly at moderate K/C ratios where the bias-variance balance is non-trivial.

Table 2: CIFAR-100 multiclass bridge. B_j^2 is computed via OVR linear heads on ResNet-18 features. The theory correctly predicts Oracle dominance ($\mathrm{SNR}_{\text{concept}} > C - 1 = 3$) for all seeds.

Seed	B_j^2	σ^2	d_{eff}	SNR	FedAvg Acc	Oracle Acc	Match
42	1.393	0.158	2560	7.22	73.8%	86.9%	✓
43	1.372	0.159	2560	11.31	75.4%	87.6%	✓
44	1.374	0.158	2560	9.85	66.3%	83.2%	✓

5.2 CIFAR-100 Bridge

To connect the theory to real data, we compute the B_j^2 proxy on CIFAR-100 with ResNet-18 features ($d = 128$).

Setup. We create 4 concepts via disjoint label splits over 20 superclasses, with $K = 10$ clients. For each concept j, we fit a one-vs-rest (OVR) linear head $W_j \in \mathbb{R}^{d \times n_{\text{classes}}}$ via regularized OLS, then compute $B_j^2 = \|W_j - \bar{W}\|_F^2$ in the Frobenius norm over the head-weight space.

Results. Table 2 shows that the theory correctly predicts Oracle dominance across all 3 seeds: the estimated $\mathrm{SNR}_{\text{concept}}$ ranges from 7.2 to 11.3, well above the threshold $C - 1 = 3$. The Oracle achieves 83–88% accuracy versus 66–75% for FedAvg, a 13–17 percentage point advantage consistent with the large B_j^2.

5.3 Regime Boundary Analysis

We verify the crossover boundary by tracking per-round FedAvg vs. Oracle dominance across 8 configurations varying K/C and δ (Figure 5).

When $\delta = 0.5$ ($\mathrm{SNR}_{\text{concept}} \approx 10$, above $C-1 = 3$ but with only $K/C = 2$ clients per concept), FedAvg still dominates in 36/40 rounds due to the high variance of concept-level estimation with few clients. This illustrates that the effective $\mathrm{SNR}_{\text{concept}}$ depends not only on δ but on the sample-size-adjusted ratio.

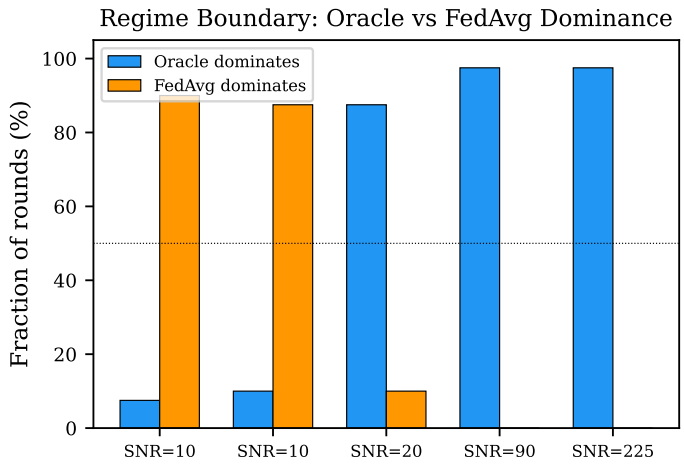

Figure 5: Regime boundary analysis. When estimated $\mathrm{SNR}_{\mathrm{concept}}$ is low (e.g., $K = 8$, $\delta = 0.5$), FedAvg dominates the majority of rounds. When $\mathrm{SNR}_{\mathrm{concept}}$ is high, Oracle dominates. The transition aligns with the theoretical boundary $\mathrm{SNR}_{\mathrm{concept}} = C - 1$.

6 Related Work

Clustered federated learning. IFCA [Ghosh et al., 2020] alternates between cluster assignment (by loss) and per-cluster model updates, with convergence guarantees assuming sufficient separation. CFL [Sattler et al., 2021] clusters clients by gradient cosine similarity. FeSEM [Ghosh et al., 2020] uses stochastic EM. FedRC and related methods [Chen et al., 2024] extend clustering to handle concept drift. These works focus on *how to cluster*, assuming clustering is beneficial. We address the complementary question of *when to cluster*.

Personalized FL. APFL [Deng et al., 2020] interpolates between local and global models with a fixed coefficient. Ditto [Li et al., 2021] regularizes toward the global model. pFedMe [T Dinh et al., 2020] uses Moreau envelopes. These methods use heuristic mixing without characterizing the optimal interpolation as a function of problem parameters. Our shrinkage estimator provides the principled, data-adaptive alternative.

FL under concept drift. FedDrift [Jothimurugesan et al., 2023] detects drift and creates new cluster models. Flash [Panchal et al., 2023] uses attention-based adaptation. FedCCFA [Chen et al., 2024] combines clustering with drift adaptation. These methods propose specific algorithms without characterizing when concept-level treatment is worth the variance cost.

Statistical theory for FL. Ghosh et al. [2020] analyze IFCA's convergence rate under separation conditions. Recent works [Xu et al., 2024, Kim et al., 2024] study convergence rates for personalized FL. Our contribution is orthogonal: we characterize the *crossover phenomenon* itself, providing both an upper bound (shrinkage) and a matching lower bound (minimax).

7 Conclusion

We have shown that the aggregation granularity question in federated learning under concept drift has a precise answer: concept-level aggregation outperforms global aggregation if and only if the concept signal-to-noise ratio exceeds $C - 1$. This crossover is fundamental (minimax lower bounds), and a James-Stein shrinkage estimator provides a near-optimal data-adaptive strategy.

Limitations. Our theory assumes Gaussian linear regression with balanced concepts and known concept assignments. Extending to nonlinear models (where d is replaced by an effective dimension), unbalanced

concepts, and unknown assignments with noisy concept inference are important future directions. Our real-data validation uses a single benchmark (CIFAR-100 with frozen features); additional datasets would strengthen the empirical bridge.

References

Hong-You Chen et al. Addressing concept shift in online federated learning. In *NeurIPS*, 2024.

Yuyang Deng, Mohammad Mahdi Kamani, and Mehrdad Mahdavi. Adaptive personalized federated learning. *arXiv preprint arXiv:2003.13461*, 2020.

Bradley Efron and Carl Morris. Stein's estimation rule and its competitors—an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341):117–130, 1973.

Avishek Ghosh, Jichan Chung, Dong Yin, and Kannan Ramchandran. An efficient framework for clustered federated learning. In *NeurIPS*, 2020.

William James and Charles Stein. Estimation with quadratic loss. *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, 1:361–379, 1961.

Ellango Jothimurugesan, Kevin Hsieh, Jianyu Wang, Gauri Joshi, and Phillip B Gibbons. Federated learning under distributed concept drift. In *AISTATS*, 2023.

Yoojin Kim et al. Communication-efficient personalized federated learning for speech-to-intent classification. In *ICML*, 2024.

Tian Li, Shengyuan Hu, Ahmad Beirami, and Virginia Smith. Ditto: Fair and robust federated learning through personalization. In *ICML*, 2021.

Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *AISTATS*, 2017.

Kunjal Panchal, Sunav Choudhary, Subrata Mitra, Koyel Mukherjee, Somdeb Sarkhel, Saayan Mitra, and Hui Guan. Flash: Concept drift adaptation in federated learning. In *ICML*, 2023.

Felix Sattler, Klaus-Robert Müller, and Wojciech Samek. Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints. *IEEE TNNLS*, 2021.

Canh T Dinh, Nguyen Tran, and Josh Nguyen. Personalized federated learning with Moreau envelopes. In *NeurIPS*, 2020.

Alexandre B Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.

Rui Xu et al. Personalized federated learning with inferred collaboration graphs. In *ICML*, 2024.

A Proof of Proposition 1

The global OLS estimator converges to the population minimizer of the mixture $\frac{1}{C}\sum_j P_j$. Because each P_j has a different optimal predictor w_j^*, the population minimizer targets $\bar{w}^* = \frac{1}{C}\sum_j w_j^*$, which is biased away from any individual w_j^* by $B_j^2 = \|w_j^* - \bar{w}^*\|_\Sigma^2$. The variance term follows from standard OLS theory with Kn effective samples. The concept-level estimator targets w_j^* directly (zero bias) with $(K/C)n$ samples, giving the stated variance.

B Proof of Theorem 3

Lower bound (5): Consider $C = 2$ with $w_1^* = \bar{w}^* + (\Delta/2)e_1$ and $w_2^* = \bar{w}^* - (\Delta/2)e_1$. Any estimator not using concept labels must target a single $\hat{w}$. By the triangle inequality, $\|\hat{w} - w_1^*\|_\Sigma^2 + \|\hat{w} - w_2^*\|_\Sigma^2 \geq \Delta^2/2$.

Lower bound (6): This is the standard minimax rate for d-dimensional linear regression with $(K/C)n$ Gaussian samples [Tsybakov, 2009].

C Extended Experimental Results

Table 3 provides results for selected configurations from the full 108-setting sweep.

Table 3: Extended results (selected configurations). All MSE values are seed-averaged.

K	C	δ	τ	SNR	FedAvg	Oracle	Shrinkage	IFCA	CFL	APFL
10	2	0.3	∞	1.1	1.067	1.091	1.067	1.091	1.143	**1.065**
10	4	1.0	15	12.5	1.197	1.159	**1.155**	1.159	1.218	1.173
10	4	3.0	∞	113	3.657	**1.159**	1.159	1.159	1.225	1.286
20	8	1.0	15	12.5	1.235	1.231	**1.158**	1.231	1.303	1.224
40	4	3.0	5	450	3.694	1.081	**1.081**	1.081	1.114	1.159
40	8	8.0	15	1600	25.74	**1.066**	1.066	1.066	1.072	1.116

D Baseline Implementation Details

IFCA. Each client evaluates the MSE of all C cluster heads on its local data, selects the best, trains locally, and uploads. The server averages uploads per cluster. This is a linear-Gaussian analogue of Ghosh et al. [2020]; the original IFCA uses gradient descent, which we replace with closed-form OLS.

CFL. After one round of global OLS, we compute pairwise cosine similarity of client gradient updates $\nabla_k = \hat{w}_k^{\text{local}} - \hat{w}^{\text{global}}$. Clients are clustered hierarchically with C clusters. Each client is served by its cluster's averaged model. This is a linear-Gaussian analogue of Sattler et al. [2021].

APFL. For each $\alpha \in \{0.2, 0.5, 0.8\}$, we compute $\hat{w}_k^{\text{apfl}} = \alpha\hat{w}_k^{\text{local}} + (1-\alpha)\hat{w}^{\text{global}}$ and evaluate MSE. We report the best α per configuration. This is a tuned linear-Gaussian analogue of Deng et al. [2020].